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ORIGINAL PAPER

Inferential procedures based on the integrated empirical characteristic function

Sangyeol Lee[1] · Simos G. Meintanis[2,3] · Minyoung Jo[1]

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Abstract
We introduce the novel notion of the integrated characteristic function and its empirical counterpart. Some basic properties of these new objects are mentioned and in turn utilized in order to construct new procedures for testing goodness of fit to parametric distributions, for testing symmetry and homogeneity, and for testing independence. Asymptotic results are obtained, while corresponding Monte Carlo results on the finite-sample behavior of the procedures are also included.

Keywords Characteristic function · Empirical characteristic function · Goodness of fit · Symmetry test · Two-sample test · Independence test

Mathematics Subject Classification 62G10 · 62G20

1 Introduction

Typically, goodness-of-fit (GOF) procedures utilize the distribution function (DF) and its empirical counterpart, the empirical DF. These procedures include the Kolmogorov–Smirnov, Cramér–von Mises and Anderson–Darling which form, sort of speak, the 'Holly Trinity' of such tests. The interested reader is referred to Thas (2010) for a book length treatment of the corresponding theory and applications. Although the empirical DF is still the standard approach to GOF testing, recently there has been

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✉ Simos G. Meintanis
simosmei@econ.uoa.gr

1 Department of Statistics, Seoul National University, Seoul, South Korea

2 Department of Economics, National and Kapodistrian University of Athens, Athens, Greece

3 Unit for Business Mathematics and Informatics, North–West University, Potchefstroom, South Africa

some development of procedures based on the characteristic function (CF) and the empirical characteristic function (ECF). In fact, the literature has come to be quite rich and we refer to the review paper of Meintanis (2016) and the references therein for up-to-date information on ECF-based procedures. For earlier review material on the ECF, we refer to the papers of Csörgő (1984), Epps (2005), Hušková and Meintanis (2008a, b) and to the book by Ushakov (1999).

While the empirical DF certainly incorporates all the information about the underlying DF and thus constitutes a standard tool for GOF testing, the corresponding integrated quantity, the integrated empirical DF has also been used for the same purpose; see Klar (1999, 2001) and Henze and Nikitin (2000, 2003). In this connection we note that the integrated DF is in one-to-one correspondence with the underlying law, see Müller (1996), and hence, this quantity is at least as informative as the conventional DF. In fact, the integrated DF has a relatively long history dating back to De Groot (1970), and it is routinely used in the actuarial literature. For more details on the integrated DF as well as other more general so-called stop-loss transforms, the reader is referred to the volume by Kass et al. (2002), §7.6.

The purpose of this paper is to propose analogous integrated quantities in the Fourier domain, namely the integrated CF and integrated ECF, and use these quantities for statistical inference. The remainder of this work is outlined as follows. In Sect. 2 the integrated quantities are introduced and some of their properties are presented. In Sect. 3 we suggest L2-type goodness-of-fit test procedures which involve the integrated ECF. Our prime applications are GOF tests for parametric families of distributions, symmetry testing, and testing for the two-sample problem and independence. In Sect. 4 we study the asymptotic properties of the procedures. Corresponding finite-sample results are included in Sect. 5, and we conclude with Sect. 6. All proofs are collected in "Appendix."

2 Integrated characteristic function

Let $X_1, \ldots, X_n$, be independent copies of an arbitrary random variable X and write $F(x)$ and $\varphi(t)$ for the corresponding DF and CF, respectively. We define the empirical characteristic function (ECF) as

$$\varphi_n(t) = \frac{1}{n}\sum_{j=1}^{n} \mathrm{e}^{itX_j}, \quad i = \sqrt{-1}. \tag{2.1}$$

L2-type GOF procedures based on the ECF $\varphi_n(t)$ are often formulated as

$$\int_{-\infty}^{\infty} |\varphi_n(t) - \varphi(t)|^2 w(t)\mathrm{d}t, \tag{2.2}$$

where $w(t) = w(-t) > 0$ is a continuous weight function such that $\int_{-\infty}^{\infty} w(t)\mathrm{d}t < \infty$, which is introduced in order to smooth the periodic component which is always present in the ECF (and sometimes present even in the CF itself). While the ECF and test

statistics such as that in (2.2) have become something of a standard in GOF testing, recently there has also been an attempt to modify the ECF for different purposes. Specifically Meintanis et al. (2014) and Meintanis and Ushakov (2016) suggested the probability weighted ECF which is introduced for the purpose of avoiding the use of a weight function in (2.2), a choice which is often made on an ad hoc basis. This approach has its roots in the classical inference based on the so-called probability weighted moments. Here we follow the idea in the earlier work by Klar (1999, 2001) and Henze and Nikitin (2000, 2003) on the integrated empirical DF in defining an analogous quantity in the Fourier domain.

To this end, notice that the ECF and CF satisfy $\varphi_n(-t) = \overline{\varphi_n(t)}$ and $\varphi(-t) = \overline{\varphi(t)}$, where $\overline{z}$ denotes the conjugate of a complex number z, and hence, all the information regarding these quantities is already contained in the positive axis $t > 0$. Therefore, the integrated ECF (IECF) is defined as

$$\Phi_{n,w}(t) = \int_0^t \varphi_n(u)w(u)\mathrm{d}u := C_{n,w}(t) + iS_{n,w}(t), \tag{2.3}$$

where $C_{n,w}(\cdot)$ (resp. $S_{n,w}(\cdot)$) denotes the real part (resp. imaginary part) of the IECF. Since $|\varphi_n(t)| \leq 1$ and $\int_{-\infty}^{\infty} w(t)\mathrm{d}t < \infty$, we have by Fubini's theorem and the unbiasedness of the ECF,

$$\begin{aligned}\mathbb{E}\left(\Phi_{n,w}(t)\right) &= \mathbb{E}\left(\int_0^t \varphi_n(u)w(u)\mathrm{d}u\right) = \int_0^t \mathbb{E}\left(\varphi_n(u)\right)w(u)\mathrm{d}u \\ &= \int_0^t \varphi(u)w(u)\mathrm{d}u := \Phi_w(t),\end{aligned} \tag{2.4}$$

and therefore we shall call $\Phi_w(t)$, the integrated characteristic function (ICF).

We may also define the joint IECF as

$$\Phi_{n,w}(t_1, t_2) = \int_0^{t_1}\int_0^{t_2} \varphi_n(u_1, u_2)w(u_1)w(u_2)\mathrm{d}u_2\mathrm{d}u_1, \tag{2.5}$$

where

$$\varphi_n(t_1, t_2) = \frac{1}{n}\sum_{j=1}^{n} \mathrm{e}^{i(t_1X_j + t_2Y_j)}, \tag{2.6}$$

is the joint ECF computed from independent copies (X_j, Y_j), $j = 1, \ldots, n$, of an arbitrary random vector (X, Y). The joint IECF is an unbiased estimate of its population counterpart, the joint ICF, which is defined as

$$\Phi_w(t_1, t_2) = \int_0^{t_1}\int_0^{t_2} \varphi(u_1, u_2)w(u_1)w(u_2)\mathrm{d}u_2\mathrm{d}u_1. \tag{2.7}$$

Analogous definitions may be stated for the joint ICF for random variables in general dimension greater than two.

The ICF has certain properties which directly relate to corresponding properties of the CF. Specifically and in obvious notation, we have:

Proposition 2.1 *Uniqueness:* $\Phi_{X,w}(t) = \Phi_{Y,w}(t), \forall t > 0$, *if and only if* $F_X \equiv F_Y$.

Proof Deferred to "Appendix." □

Proposition 2.2 *Symmetry characterization:* $\Phi_w(t)$ *is real* $\forall t > 0$, *if and only if* F *is symmetric around zero.*

Proof Deferred to "Appendix." □

Proposition 2.3 *Independence characterization: The random variables* X *and* Y *are independent if and only if*

$$\Phi_w(t_1, t_2) = \Phi_{X,w}(t_1)\Phi_{Y,w}(t_2), \tag{2.8}$$

$\forall t_1, t_2 \in \mathbb{R}$.

Proof Deferred to "Appendix." □

Notice that unlike Propositions 2.1 and 2.2 where we only consider $t > 0$, in Proposition 3 we require (2.8) to be true not just for $t_1, t_2 > 0$, but for all real t_1, t_2. The reason will become apparent in the proofs of the propositions provided in "Appendix." We simply mention here as a consequence that problems, such as the independence problem, which are inherently multivariate formally require the ICF to result by integrating the corresponding CF over the full domain of its definition, and not merely over the positive quadrant. Since, however, the associated computations are considerably more complicated, in what follows we work with the joint IECF and ICF as defined by (2.5) and (2.7), respectively. It should be kept in mind that this compromise does not affect at all the inference procedures suggested in Sects. 3.1–3.3, but only those in Sect. 3.4. Even in this case though any loss of generality essentially hinges upon such rare phenomena as CFs that coincide over infinite intervals but correspond to different distributions. Although 'pathological' cases as these are occasionally reported in the literature, see Lucas (1970, p. 85), Ushakov (1999, p. 264) and Gneiting (2001), they should be viewed as curiosities of purely academic interest and as such do not affect the soundness of the suggested procedure, at least for all practical purposes.

3 Goodness of fit based on the IECF

3.1 Goodness of fit for parametric distributions

We begin by considering the composite null hypothesis

$$H_0 : F \in \mathcal{F}_{\delta,c} \text{ for some } \delta \in \mathbb{R}, c > 0, \tag{3.1}$$

where $\mathcal{F}_{\delta,c}$ denotes an arbitrary location-scale family of distributions with DF $F_{\delta,c}(x) = F_0(\frac{x-\delta}{c})$, with δ and c denoting the location and scale parameter, respectively. For simplicity F_0 is assumed to be symmetric around zero, and it contains no

further parameters other than δ and c. In this context we propose to use the discrepancy measure

$$D_{n,w}(t) = \left|\widehat{\Phi}_{n,w}(t) - \Phi_{0,w}(t)\right|^2, \tag{3.2}$$

where $\Phi_{0,w}(t)$ denotes the ICF corresponding to F_0 and $\widehat{\Phi}_{n,w}(t)$ is the IECF computed from the standardized data

$$Z_j = \frac{X_j - \widehat{\delta}_n}{\widehat{c}_n}, \quad j = 1, \ldots, n, \tag{3.3}$$

by using suitable estimators $(\widehat{\delta}_n, \widehat{c}_n)$ of (δ, c). Typically consistent estimators are used. However, in order to entertain the full spectrum of favorable properties, asymptotic as well as finite-sample, of the test statistics we will impose further requirements on these estimators; refer to Sects. 4 and 5.1.

In view of the preceding reasoning, the natural test statistic would be

$$\int_0^\infty D_{n,w}(t) W(t) \mathrm{d}t, \tag{3.4}$$

where $W(t)$ is another weight function with analogous properties as $w(t)$. However, it turns out that an equivalent test to that in (3.4) may be written in a closed form that often avoids numerical integration. To this end write from (3.2),

$$D_{n,w}(t) = \left|\widehat{\Phi}_{n,w}(t)\right|^2 + \Phi_{0,w}^2(t) - 2\widehat{C}_{n,w}(t)\Phi_{0,w}(t), \tag{3.5}$$

where $\widehat{C}_{n,w}(t)$ denotes the real part of $\widehat{\Phi}_{n,w}(t)$, and notice that $\Phi_{0,w}^2(t)$ does not depend on the observations. Hence, we suggest the test statistic

$$T_{n,W,w} = \int_0^\infty \left(\left|\widehat{\Phi}_{n,w}(t)\right|^2 - 2\widehat{C}_{n,w}(t)\Phi_{0,w}(t)\right) W(t)\mathrm{d}t. \tag{3.6}$$

In turn by straightforward calculations, we have that

$$T_{n,W,w} = \frac{1}{n^2}\sum_{j,k=1}^{n} I_{jk} - \frac{2}{n}\sum_{j=1}^{n} I_j, \tag{3.7}$$

where $I_{jk} = I_{W,w}(Z_j, Z_k)$ and $I_j = I_{W,w}(Z_j)$ with

$$I_{W,w}(x, y) = \int_0^\infty I_w(t; x, y) W(t)\mathrm{d}t, \quad I_{W,w}(x) = \int_0^\infty I_w(t; x)\Phi_{0,w}(t) W(t)\mathrm{d}t,$$

and

$$I_w(t; x, y) = \int_0^t \int_0^t \cos(ux - vy) w(u) w(v)\mathrm{d}u\mathrm{d}v, \quad I_w(t; x) = \int_0^t \cos(ux) w(u)\mathrm{d}u.$$

Clearly the integral $I_{W,w}(x, y)$ only depends on the weight functions and not on the null hypothesis H_0 being tested. Hence, the test statistic is further advanced by considering weight functions that simplify calculations. To this end and while the two weight functions, namely the weight function $w(t)$ figuring in the definition of the IECF and the weight function $W(t)$ figuring in the definition of the test statistic, can potentially be different from each other, here as well as in the next sections we take $W(t) = w(t) = \mathrm{e}^{-at},\ a > 0$. Then if we write $I_a(x, y)$ for the resulting integral by straightforward algebra we have

$$I_a(x, y) = \frac{48a^4 + x(x-y)^2 y + 2a^2\left(x^2 + y^2 + 4xy\right)}{a\left(4a^2 + x^2\right)\left(9a^2 + (x-y)^2\right)\left(4a^2 + y^2\right)}. \tag{3.8}$$

Next we specify the procedure for certain distributions under test:
Normal distribution By using the aforementioned weight function, we are led to $\Phi_{0,w}(t) := \Phi_{0,a}(t)$ and $I_{W,w}(x) := I_a(x)$ where

$$\Phi_{0,a}(t) = \sqrt{\frac{\pi}{2}}\mathrm{e}^{a^2/2}\left(\mathrm{Erf}\left[\frac{a+t}{\sqrt{2}}\right] - \mathrm{Erf}\left[\frac{a}{\sqrt{2}}\right]\right),$$

and

$$\begin{aligned} I_a(x) = a\int_0^\infty \mathrm{Erf}\left(\frac{a+t}{\sqrt{2}}\right)\mathrm{e}^{-at}\mathrm{d}t - \mathrm{Erf}\left(\frac{a}{\sqrt{2}}\right) \\ - \int_0^\infty \mathrm{Erf}\left(\frac{a+t}{\sqrt{2}}\right)(a\cos tx - x\sin tx)\mathrm{e}^{-2at}\mathrm{d}t + \mathrm{Erf}\left(\frac{a}{\sqrt{2}}\right)\frac{2a^2 - x^2}{4a^2 + x^2}, \end{aligned}$$

with $\mathrm{Erf}(x) = (2/\sqrt{\pi})\int_0^x \mathrm{e}^{-u^2}\mathrm{d}u$ being the error function.
Cauchy distribution Clearly

$$\Phi_{0,a}(t) = \frac{1}{1+a}\left(1 - \mathrm{e}^{-(1+a)t}\right).$$

Also by tedious but otherwise straightforward calculations, we get

$$I_a(x) = \frac{14a^3 + 24a^4 + 14ax^2 + x^2\left(2 + x^2\right) + a^2\left(2 + 25x^2\right)}{\left(a^2 + x^2\right)(1+2a)\left(4a^2 + x^2\right)\left(1 + 6a + 9a^2 + x^2\right)}.$$

Remark 3.1 While the null hypothesis in (3.1) is stated in terms of symmetric location-scale families, this is chosen for the shake of convenience of exposition. In other words, an IECF goodness-of-fit statistic with the discrepancy measure figuring in (3.2) applies to any given family of distributions under test. Nevertheless, this more general setting with possibly non-symmetric distributions requires, not (3.5), but the full expansion of the expression (3.2) as

$$D_{n,w}(t) = \left|\Phi_{n,w}(t)\right|^2 + \left|\Phi_{0,w}(t)\right|^2 - 2\left(C_{n,w}(t)C_{0,w}(t) + S_{n,w}(t)S_{0,w}(t)\right),$$

involving the real as well as the imaginary part of the IECF, $C_{n,w}(\cdot)$ and $S_{n,w}(\cdot)$, respectively, defined in (2.3), and their corresponding population counterparts under the null hypothesis, $C_{0,w}(\cdot)$ and $S_{0,w}(\cdot)$. Moreover, if the distribution under test involves a parameter other than location or scale, such as a shape parameter, this parameter will also have to be estimated, and then both $C_{0,w}(\cdot)$ and $S_{0,w}(\cdot)$ as well as $|\Phi_{0,w}|^2 = C_{0,w}^2 + S_{0,w}^2$, will be data dependent in that they involve the estimate of this shape parameter. Hence, the test statistic in (3.6) will involve some extra terms, but will be otherwise of the same type.

3.2 Testing for symmetry

We consider the null hypothesis

$$\mathcal{H}_0 : F \text{ is symmetric around } \delta, \text{ for some } \delta \in \mathbb{R}. \tag{3.9}$$

In view of Proposition 2.2 we suggest the test statistic

$$\mathcal{T}_{n,W,w} = \int_0^\infty \widehat{S}_{n,w}^2(t) W(t) \mathrm{d}t, \tag{3.10}$$

where $\widehat{S}_{n,w}(t)$ denotes the imaginary part of the IECF corresponding to the data $Y_j = X_j - \widehat{\delta}_n,\ j = 1, \ldots, n$, with $\widehat{\delta}_n$ being a suitable estimator of δ (such as the sample median for example). In turn by straightforward calculations we have that

$$\mathcal{T}_{n,W,w} = \frac{1}{2n^2} \sum_{j,k=1}^{n} \left(I_{jk}^{(-)} - I_{jk}^{(+)} \right), \tag{3.11}$$

where $I_{jk}^{\pm} = I_{W,w}^{\pm}(Y_j, Y_k)$ with

$$I_{W,w}^{(\pm)}(x, y) = \int_0^\infty I_w^{(\pm)}(t; x, y) W(t) \mathrm{d}t,$$

with

$$I_w^{(\pm)}(t; x, y) = \int_0^t \int_0^t \cos(ux \pm vy) w(u) w(v) \mathrm{d}u \mathrm{d}v.$$

If we let $W(t) = w(t) = \mathrm{e}^{-at},\ a > 0$, the integral $I_a^{(-)}(x, y)$, say, already appears as $I_a(x, y)$ in (3.8). Likewise we have

$$I_a^{(+)}(x, y) = \frac{48a^4 - x(x+y)^2 y + 2a^2 \left(x^2 + y^2 - 4xy\right)}{a \left(4a^2 + x^2\right) \left(9a^2 + (x+y)^2\right) \left(4a^2 + y^2\right)},$$

which finally renders the symmetry test statistic in closed form free of numerical integration.

3.3 Testing for the two-sample problem

Suppose that in addition to $X_1, \ldots, X_n$, we are given independent copies $Y_1, \ldots, Y_m$, of the random variable Y, and for simplicity take $n = m$. We wish to test the null hypothesis

$$\mathbb{H}_0 : F_X \equiv F_Y. \tag{3.12}$$

Then in view of Proposition 2.1, we suggest to reject the null hypothesis $\mathbb{H}_0$ for large values of the test statistic

$$\Psi_{n,W,w} = \int_0^\infty \left|\Phi_{X,n,w}(t) - \Phi_{Y,n,w}(t)\right|^2 W(t)\mathrm{d}t, \tag{3.13}$$

where $\Phi_{X,n,w}(t)$ and $\Phi_{Y,n,w}(t)$ are the IECFs corresponding to $\{X_j\}_{j=1}^n$ and $\{Y_j\}_{j=1}^n$, respectively.

If we take $W(t) = w(t) = \mathrm{e}^{-at}$ as weight function and write $\Psi_{n,a}$ for the resulting test statistic, then following some computations using standard trigonometric formulae we obtain from (3.13),

$$\Psi_{n,a} = \frac{1}{n^2} \sum_{j,k=1}^{n} \left\{ H\left(X_j, X_k\right) + H\left(Y_j, Y_k\right) - H\left(X_j, Y_k\right) - H\left(X_k, Y_j\right) \right\}, \tag{3.14}$$

where

$$H(x, y) = \frac{1}{\left(a^2 + x^2\right)\left(a^2 + y^2\right)} \left\{ \frac{3a^3 + 5axy - ax^2 - ay^2}{9a^2 + (x - y)^2} + \frac{ax^2 - a^3 - 3axy}{4a^2 + x^2} + \frac{ay^2 - a^3 - 3axy}{4a^2 + y^2} + a + \frac{xy}{a} \right\}.$$

3.4 Testing for independence

Proceeding in an analogous manner as in the previous two subsections, and in view of Proposition 2.3, we suggest to reject the null hypothesis

$$\eta_0 : X \text{ and } Y \text{ are independent}, \tag{3.15}$$

for large values of the test statistic

$$V_{n,W,w} = \int_0^\infty \int_0^\infty \left|\Phi_{n,w}(t_1, t_2) - \Phi_{X,n,w}(t_1)\Phi_{Y,n,w}(t_2)\right|^2 W(t_1)W(t_2)\mathrm{d}t_1\mathrm{d}t_2. \tag{3.16}$$

Setting $W(t) = w(t) = \mathrm{e}^{-at}$, and writing $V_{n,a}$ for the resulting test statistic, we have following straightforward calculations

$$V_{n,a} = \int_0^\infty \int_0^\infty \left\{ \left(\frac{1}{n} \sum_{j=1}^n A_1 - \frac{1}{n^2} \sum_{k,l=1}^n B_1 \right)^2 + \left(\frac{1}{n} \sum_{j=1}^n A_2 - \frac{1}{n^2} \sum_{k,l}^n B_2 \right)^2 \right\} \mathrm{e}^{-a(t_1+t_2)} \mathrm{d}t_1 \mathrm{d}t_2, \tag{3.17}$$

where

$$A = A\,(t_1, t_2) := \int_0^{t_1} \mathrm{e}^{iuX_j} \mathrm{e}^{-au} \mathrm{d}u \int_0^{t_2} \mathrm{e}^{ivY_j} \mathrm{e}^{-av} \mathrm{d}v := A_1 + iA_2$$

and

$$B = B(t_1, t_2) := \int_0^{t_1} \mathrm{e}^{iuX_k} \mathrm{e}^{-au} \mathrm{d}u \int_0^{t_2} \mathrm{e}^{ivY_l} \mathrm{e}^{-av} \mathrm{d}v := B_1 + iB_2$$

with

$$\begin{aligned}
A_1 &= f_a(X_j, t_1) f_a(Y_j, t_2) - g_a(X_j, t_1) g_a(Y_j, t_2);\\
A_2 &= f_a(X_j, t_1) g_a(Y_j, t_2) + g_a(X_j, t_1) f_a(Y_j, t_2);\\
B_1 &= f_a(X_k, t_1) f_a(Y_l, t_2) - g_a(X_k, t_1) g_a(Y_l, t_2);\\
B_2 &= f_a(X_k, t_1) g_a(Y_l, t_2) + g_a(X_k, t_1) f_a(Y_l, t_2);\\
f_a(x, t) &= \frac{x \sin(tx)\mathrm{e}^{-at} - a\cos(tx)\mathrm{e}^{-at} + a}{x^2 + a^2};\\
g_a(x, t) &= \frac{-x \cos(tx)\mathrm{e}^{-at} - a\sin(tx)\mathrm{e}^{-at} + x}{x^2 + a^2}.
\end{aligned}$$

In an actual calculation of $V_{n,a}$, one can also use Monte Carlo simulations based on iid exponential random samples, because it is hard to get analytically. For example, in the calculation of the integral $\int_\alpha^\beta g(u)\mathrm{e}^{-au}\mathrm{d}u$ for real-valued functions g, we use the approximation: $\sum_{j=1}^N g(z_j) I(\alpha < z_j < \beta)/a$ by generating iid exponential random variables z_j with intensity $a > 0$, $j = 1, \ldots, N$, for a large integer N. Here we use $N = 200$.

We close this section by noting that the proposed procedures are analogous to those suggested for the conventional ECF. In this connection ECF methods closely related to ours may be found in Epps (2005), Meintanis and Swanepoel (2007), Henze et al. (2003), Meintanis (2005), Alba Fernández et al. (2008), Hušková and Meintanis (2008c), and Ngatchou-Wandji (2009), among others. For a comprehensive bibliography of ECF-related inference procedures, the reader is referred to the review paper of Meintanis (2016), while some extra entries include Meintanis and Hlávka (2010), Balakrishnan et al. (2013), Székely and Rizzo (2013), Fan et al. (2017), Baringhaus et al. (2017) and Francq et al. (2017). However, while these authors employ the ordinary ECF, here we utilize the IECF.

4 Asymptotics

4.1 Goodness-of-fit test

Let $\theta = (\delta, c)$ and θ_0 be true parameter vector. Let $\hat{\varphi}_n$ be the ECF based on $Z_j = (X_j - \hat{\delta}_n)/\hat{c}_n$, $j = 1, \ldots, n$, and let $\varphi_0(t)$ be the CF of F_0. The asymptotic distribution of $T_{n,W,w}$ is obtained under the following assumption:

(**A1**) Let $\hat{\theta}_n$ be the estimator of θ_0 such that under H_0,

$$\sqrt{n}\left(\hat{\theta}_n - \theta_0\right) = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \mathbf{l}_j(\theta_0) + o_P(1),$$

where $\mathbf{l}_j(\theta_0)$ are two-dimensional iid random vectors with $E_{\theta_0}\mathbf{l}_1(\theta_0) = 0$ and $E_{\theta_0}\mathbf{l}_1^T(\theta_0)\mathbf{l}_1(\theta_0) < \infty$.

More precisely, we have the following: see "Appendix" for the proof.

Theorem 4.1 *Under H_0 and (**A1**),*

$$nT_{n,W,w} \xrightarrow{d} T_{W,w} := \int_0^\infty \left\{ \int_0^t \mathcal{Z}_1(u) w(u) \mathrm{d}u \right\}^2 W(t)\mathrm{d}t - \int_0^\infty \Phi_{0,w}^2(t) W(t)\mathrm{d}t,$$

where $\{\mathcal{Z}_1(t) : t \in \mathbb{R}\}$ is a mean zero Gaussian process with covariance being the same as $\omega(s, t)$ in Remark 3.3 of Meintanis and Swanepoel (2007).

4.2 Symmetry test

Let δ_0 be the true parameter under $\mathcal{H}_0$ and let $\hat{\varphi}_n$ be the ECF based on $Z_j = X_j - \hat{\delta}_n$, $j = 1, \ldots, n$. Then, under the assumption:

(**A2**) Let $\hat{\delta}_n$ be the estimator of δ_0 such that under $\mathcal{H}_0$,

$$\sqrt{n}\left(\hat{\delta}_n - \delta_0\right) = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \mathbf{l}_j(\delta_0) + o_P(1),$$

where $\mathbf{l}_j(\delta_0)$ are iid random variables with $E_{\delta_0}\mathbf{l}_1(\delta_0) = 0$ and $E_{\delta_0}\mathbf{l}_1^2(\delta_0) < \infty$, we have the following, the proof of which is essentially the same as that of Theorem 4.1 and is omitted for brevity.

Theorem 4.2 *Under $\mathcal{H}_0$ and (**A2**),*

$$n\mathcal{T}_{n,W,w} \xrightarrow{d} \mathcal{T}_{W,w} := \int_0^\infty \left\{ \int_0^t \mathrm{Im}\{\mathcal{Z}_1(u)\} w(u)\mathrm{d}u \right\}^2 W(t)\mathrm{d}t.$$

where $\{\mathcal{Z}_1(t) : t \in \mathbb{R}\}$ is a zero-mean Gaussian process with covariance being the same as $\omega(s, t)$ in Remark 3.3 of Meintanis and Swanepoel (2007).

4.3 Two-sample test

Let $\hat{\varphi}_{X,n}$ and $\hat{\varphi}_{Y,n}(t)$ be the ECF based on $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$, respectively. Then, we have the following, the proof of which is essentially the same as that of Theorem 4.1 and is omitted for brevity.

Theorem 4.3 *Under* $\mathbb{H}_0$,

$$n\Psi_{n,a} \xrightarrow{d} \Psi_a := \int_0^\infty \left| \int_0^t \mathcal{Z}_2(u) w(u) \mathrm{d}u \right|^2 w(t) \mathrm{d}t.$$

where $\{\mathcal{Z}_2(t) : t \in \mathbb{R}\}$ is a zero-mean Gaussian process with covariance being the same as $K(s,t)$ in Theorem 2.1 of Meintanis (2005).

4.4 Independence test

Let $\hat{\varphi}_{X,n}$ and $\hat{\varphi}_{Y,n}(t)$ be the ECF based on $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$, respectively, and let $\hat{\varphi}_{X,Y,n}$ be the joint ECF based on $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$. Then, we have the following, the proof of which is essentially the same as that of Theorem 4.1 and is omitted for brevity.

Theorem 4.4 *Under* η_0,

$$nV_{n,a} \xrightarrow{d} V_a := \int_0^\infty \int_0^\infty \left| \int_0^{t_2} \int_0^{t_1} \mathcal{Z}_3(u,v) w(u) w(v) \mathrm{d}u \mathrm{d}v \right|^2 w(t_1) w(t_2) \mathrm{d}t_1 \mathrm{d}t_2,$$

where $\{\mathcal{Z}_3(t,s) : t, s \in \mathbb{R}\}$ is a zero-mean Gaussian process with covariance being the same as that of ζ in Székely et al. (2007), p. 2780.

5 Simulations

In this section, we study the power of the proposed tests based on IECF in comparison with other tests. In this connection we refer to the ECF tests for parametric distributions suggested by Epps (2005) and Meintanis and Swanepoel (2007), among others. The asymptotic law of such tests, although generally dependent on the true values of the parameters to be estimated, can be made free of these values with proper choice of these estimators; see Sect. 5.1. On the other hand, the tests for symmetry and homogeneity proposed by Henze et al. (2003) and Meintanis (2005), respectively, are not distribution free. Therefore, they are typically carried out by means of a certain resampling scheme which is consistent in the sense of preserving the nominal level of the test. For testing symmetry, the wild bootstrap which we employ is shown to be asymptotically valid by Neuhaus and Zhu (1998) [see also Zhu (2005)], while corresponding results for the consistency of the permutation procedure by means of which we carry out the homogeneity test may be found in Hušková and Meintanis (2008c) and the references therein. A permutation procedure was also used for the implementation of the test for independence along the lines suggested by Székely et al. (2007).

Table 1 Computation time (s)

	ECF	IECF
Parametric distribution		
$n = 10$	2390.241	5003.323
$n = 20$	5603.079	9971.259
Symmetry		
$n = 10$	1975.864	5225.292
$n = 20$	3710.178	9215.859
Two-sample		
$n_1 = n_2 = 22$	7257.395	39412.850
Independence		
$n = 10$	216.8358	6890.272

In all four cases of testing considered herein we employ the weight function $W(t) = w(t) = \mathrm{e}^{-at}$. In this we follow the example set by conventional ECF procedures which are dominated by weight functions of the type $w_{a,b}(t) = \mathrm{e}^{-at^b}$, with $b = 1, 2$. Drawing from that experience we know that the choice of the specific type of function as weight function, say $b = 1$ or $b = 2$ in $w_{a,b}(\cdot)$, is not all that important, given that different weight functions render the corresponding test statistics equally powerful but for different values of a. On the other hand, for fixed b, proper values of a seem to have some effect on the power of the test, ranging from mild to significant depending on the null hypothesis being tested. Clearly a theoretical analysis of proper values of a that potentially lead to high power would depend on the chosen direction toward which the null hypothesis is violated. A generally accepted method of power analysis could be based on the so-called Bahadur slope; see Nikitin (1995). This is a highly technical problem that requires finding the eigenvalues of an integral operator and has only been considered in the specific case of testing for normality; see Tenreiro (2009). An alternative strategy depends on a delicate case-by-case analysis of the power function against fixed alternatives, for instance, along the lines of Baringhaus et al. (2017). In view of the above discussion, and given that all goodness-of-fit tests have non-trivial power only over a small region of the space defined by the corresponding alternative hypothesis [see Janssen (2000) and Escanciano (2009)], we suggest the Monte Carlo approach in order to determine which version of the tests is reasonably powerful over a certain set of alternatives.

To this end, we present below the results of a simulation study on the power of the proposed tests against earlier tests based on the ordinary ECF. For this task, we use Intel Core i7-7700K CPU with 4.2 GHz and 64-GB RAM: see Table 1 for the computation time for the ECF and IECF tests, which reveals that the latter takes a longer time than the former, as might be anticipated. Although this study will be of help in order to gain some feeling about the relative merits of the new methods, a more complete answer to this question would require a more exhaustive search which could be the subject of a separate paper; see, for instance, Allison and Santana (2015).

5.1 Tests for parametric distributions

We consider the new IECF test with $W(t) = w(t) = \mathrm{e}^{-at}$ in (3.7), say $T_{n,a}$, for the null hypothesis H_0 in (3.1) in the case of the normal and the Cauchy distribution, and compare this test with the ECF tests in (2.2) implemented as

$$\widetilde{T}_{n,a} = \int_{-\infty}^{\infty} |\widehat{\varphi}_n(t) - \varphi_0(t)|^2 \, \mathrm{e}^{-a|t|} \mathrm{d}t,$$

where $\widehat{\varphi}_n(t)$ denotes the ECF in (2.1) corresponding to the data Z_j in (3.3), and $\varphi_0(\cdot)$ denotes the CF under H_0 with $(\delta, c) = (0, 1)$. For the normal distribution we use the mean and the standard deviation as parameters (δ, c) and employ the sample mean and the sample standard deviation as estimators of δ and c, respectively. Corresponding parameters in the case of testing for the Cauchy distributions were the median $F^{-1}(1/2)$ and the semi-interquartile range $(F^{-1}(3/4) - F^{-1}(1/4))/2$ which were estimated by replacing F by F_n (the empirical DF) in their definition. In both cases, these pairs of estimators satisfy $\widehat{\delta}_n(aX_1 + b, \ldots, aX_n + b) = a\widehat{\delta}_n(X_1, \ldots, aX_n) + b$ and $\widehat{c}_n(aX_1 + b, \ldots, aX_n + b) = a\widehat{c}_n(X_1, \ldots, aX_n)$, for each $a > 0,\ b \in \mathbb{R}$. Therefore, if implemented on the standardized data (3.3), the resulting test statistic is rendered invariant with respect to location and scale in the sense that $T_n(aX_1 + b, \ldots, aX_n + b) = T_n(X_1, \ldots, X_n)$. Hence, the finite-sample as well as the asymptotic null distribution may be approximated by simply setting $\delta = 0$ and $c = 1$; see Meintanis and Swanepoel (2007).

We obtain the empirical sizes for the standard normal and Cauchy distributions with estimated parameters and samples of size $n = 10$ and $n = 20$ at the nominal level of 0.05. The standard Cauchy distribution with fixed parameters is also considered. The repetition number and bootstrap repetition numbers are commonly 500. Table 2 shows that the empirical sizes of the ECF and IECF tests are fairly close to the nominal level of 0.05, and no significant differences are observed. The figures in the parentheses stand for the standard deviations calculated from 100 repetitions. To examine the power, we consider the normal, the Cauchy, the t distribution, the skew t, and the Gumbel distribution as alternatives. Table 3 shows that the ECF and IECF tests for normality perform similarly overall, with the one being more powerful for some alternatives while at the same time giving ground to its competitor for other alternatives.

Tables 4 and 5 show the performance of the two tests for the Cauchy distribution with fixed and with estimated parameters, respectively. The results show that while the new test is clearly more powerful in the first case (fixed parameters), standardization has a greater impact compared to the ECF test, and as a result the IECF test is now competitive only for $a = 1$.

5.2 Tests for symmetry

We consider the null hypothesis $\mathcal{H}_0$ in (3.9) with $\delta = 0$ and compare $\mathcal{T}_{n,a}$, i.e., the new IECF test in (3.11) with $W(t) = w(t) = \mathrm{e}^{-at}$, with the test statistic based on the imaginary part $\Im\varphi_n(t)$ of the ECF,

Table 2 Empirical size of the test for parametric distributions under a null normal or Cauchy distribution for weight parameter a and sample size n

Null distribution	a	ECF $n = 10$	IECF	ECF $n = 20$	IECF
Normal distribution (estimated parameters)	0.25	0.051 (0.010)	0.051(0.010)	0.052 (0.010)	0.052 (0.009)
	0.5	0.051 (0.010)	0.051 (0.010)	0.052 (0.010)	0.052 (0.009)
	1	0.050 (0.009)	0.051 (0.010)	0.052 (0.009)	0.052 (0.010)
Cauchy distribution (fixed parameters)	0.25	0.053 (0.010)	0.053 (0.009)	0.053 (0.011)	0.052(0.010)
	0.5	0.052 (0.009)	0.052 (0.009)	0.053 (0.011)	0.053 (0.010)
	1	0.052 (0.009)	0.053 (0.009)	0.053 (0.012)	0.053 (0.010)
Cauchy distribution (estimated parameters)	0.25	0.051 (0.009)	0.052(0.011)	0.052 (0.010)	0.054 (0.010)
	0.5	0.052 (0.010)	0.053 (0.010)	0.052 (0.010)	0.053 (0.010)
	1	0.052 (0.010)	0.051 (0.011)	0.052 (0.011)	0.052 (0.010)

Table 3 Empirical power of the test for parametric distributions under a null normal distribution with estimated parameters for weight parameter a and sample size n

Alternative distribution	a	ECF $n = 10$	IECF	ECF $n = 20$	IECF
$t(3)$	0.25	0.138 (0.016)	0.197 (0.019)	0.233 (0.020)	0.360 (0.024)
	0.5	0.163 (0.016)	0.220 (0.020)	0.276 (0.024)	0.410 (0.026)
	1	0.188 (0.016)	0.228 (0.019)	0.323 (0.024)	0.425 (0.026)
Skew $t(3, 2)$	0.25	0.241 (0.017)	0.182 (0.017)	0.451 (0.023)	0.316 (0.021)
	0.5	0.281 (0.018)	0.225 (0.019)	0.519 (0.022)	0.395 (0.021)
	1	0.318 (0.020)	0.25 (0.020)	0.583 (0.023)	0.449 (0.020)
Gumbel (0.5, 2)	0.25	0.104 (0.013)	0.087 (0.013)	0.167 (0.017)	0.120 (0.016)
	0.5	0.125 (0.015)	0.105 (0.015)	0.214 (0.018)	0.160 (0.018)
	1	0.150 (0.016)	0.122 (0.015)	0.271 (0.019)	0.197 (0.020)
Cauchy distribution	0.25	0.564 (0.022)	0.610 (0.024)	0.844 (0.016)	0.896 (0.015)
	0.5	0.596 (0.022)	0.626 (0.022)	0.867 (0.015)	0.910 (0.013)
	1	0.614 (0.021)	0.610 (0.023)	0.880 (0.014)	0.900 (0.013)

Table 4 Empirical power of the test for parametric distributions under a null Cauchy distribution with fixed parameters for weight parameter a and sample size n

Alternative distribution	a	ECF $n = 10$	IECF	ECF $n = 20$	IECF
$t(3)$	0.25	0.075 (0.013)	0.102 (0.013)	0.096 (0.014)	0.124 (0.014)
	0.5	0.069 (0.012)	0.107 (0.013)	0.103 (0.013)	0.162 (0.016)
	1	0.059 (0.010)	0.095 (0.012)	0.113 (0.013)	0.210 (0.020)
Skew $t(3, 2)$	0.25	0.220 (0.018)	0.188 (0.019)	0.436 (0.020)	0.371 (0.021)
	0.5	0.294 (0.021)	0.346 (0.022)	0.586 (0.019)	0.650 (0.021)
	1	0.383 (0.024)	0.520 (0.023)	0.735 (0.020)	0.861 (0.017)
Gumbel (0.5, 2)	0.25	0.036 (0.008)	0.109 (0.013)	0.025 (0.007)	0.264 (0.020)
	0.5	0.035 (0.008)	0.238 (0.018)	0.023 (0.007)	0.449 (0.020)
	1	0.035 (0.007)	0.361 (0.021)	0.024 (0.006)	0.609 (0.019)
Normal distribution	0.25	0.111 (0.015)	0.159 (0.016)	0.191 (0.018)	0.230 (0.018)
	0.5	0.109 (0.013)	0.195 (0.016)	0.239 (0.021)	0.377 (0.022)
	1	0.103 (0.013)	0.194 (0.017)	0.322 (0.021)	0.583 (0.023)

$$\widetilde{\mathcal{T}}_{n,a} = \int_{-\infty}^{\infty} [\Im\varphi_n(t)]^2 \, \mathrm{e}^{-a|t|} \mathrm{d}t = \frac{1}{2n^2} \sum_{j,k=1}^{n} \left(\frac{2a}{a^2 + \left(X_j - X_k\right)^2} - \frac{2a}{a^2 + \left(X_j + X_k\right)^2} \right).$$

As already mentioned the finite-sample as well as the asymptotic null distributions of L2-type symmetry statistics, such as $\mathcal{T}_{n,a}$ and $\widetilde{\mathcal{T}}_{n,a}$, depend on the unknown underlying law F. Therefore, we use the resampling technique of Neuhaus and Zhu (1998) and Zhu (2005), in order to approximate them. Namely instead of using X_j in the computation of the test statistics, we employ the observations $\widetilde{X}_j = e_j X_j, \ j = 1, \ldots, n$, where the variables $e_1, \ldots, e_n$, are independent and identically distributed and independent of X_j, with $\mathrm{P}(e_j = \pm 1) = 1/2, \ j = 1, \ldots, n$.

Here we consider a standard normal distribution as the null and the $N + \chi^2$ distribution as an alternative, where $N + \chi^2$ stands for the distribution of $N_1 + N_2^2 - 1$, where N_1, N_2 are iid $N(0, 1)$. The repetition number and bootstrap repetition numbers are commonly 500. Table 6 (corresponding to the case of δ estimated by the sample median $\widehat{\delta}_n$) shows that the empirical sizes and powers of the ECF and IECF tests for $n = 10$ and $n = 20$ are similar to each other. Table 7 (δ is assumed known) shows pretty much the same picture in terms of power (with a slight edge for the ECF test), but with no size distortions compared to Table 6.

5.3 Tests for the two-sample problem

In this case too, the null distribution of the test statistic depends on the unknown distribution $F = F_X = F_Y$. Therefore, we performed a permutation test as follows. Conditionally on the observed values $X_1, X_2, \ldots, X_n$, and $Y_1, Y_2, \ldots, Y_n$, of the two samples, generate a number of permutations of the grouped sample, chosen at random from all possible permutations. Calculate the value of the test statistic for each permuted sample and obtain the critical point as the critical point of the permutation distribution of the test statistic. More detailed description of the procedure can be found in Meintanis (2005). We compare our test $\Psi_{n,a}$ in (3.14) for the null hypothesis $\mathbb{H}_0$ with the test statistic

$$\begin{aligned}\widetilde{\Psi}_{n,a} &= \int_{-\infty}^{\infty} \left|\varphi_{X,n}(t) - \varphi_{Y,n}(t)\right|^2 \mathrm{e}^{-a|t|} \mathrm{d}t \\ &= \frac{2a}{n^2} \sum_{j,k=1}^{n} \left(\frac{1}{a^2 + \left(X_j - X_k\right)^2} + \frac{1}{a^2 + \left(Y_j - Y_k\right)^2} - \frac{2}{a^2 + \left(X_j - Y_k\right)^2} \right),\end{aligned}$$

which is based on the ordinary ECFs $\varphi_{X,n}(t)$ and $\varphi_{Y,n}(t)$ corresponding to the data $\{X_j\}_{j=1}^n$ and $\{Y_j\}_{j=1}^n$, respectively.

We consider the normal, exponential, gamma (with shape equal to two), beta (with parameters equal to two and three), logistic L, lognormal and uniform U distributions. All distributions are centered and scaled to have mean zero and unit variance, and 500 random samples with size $n_1 = n_2 = 22$ are generated from the same distributions, picking up 500 random permutations at each time. Tables 8, 9, 10 and 11 summarize

Table 5 Empirical power of the test for parametric distributions under a null Cauchy distribution with estimated parameters for weight parameter a and sample size n

Alternative distribution	a	ECF $n = 10$	IECF	ECF $n = 20$	IECF
$t(3)$	0.25	0.046 (0.009)	0.025 (0.007)	0.065 (0.012)	0.029 (0.008)
	0.5	0.043 (0.009)	0.025 (0.007)	0.070 (0.012)	0.031 (0.008)
	1	0.034 (0.008)	0.065 (0.011)	0.068 (0.012)	0.134 (0.015)
Skew $t(3, 2)$	0.25	0.080 (0.011)	0.053 (0.010)	0.169 (0.017)	0.098 (0.013)
	0.5	0.083 (0.013)	0.066 (0.012)	0.212 (0.018)	0.160 (0.018)
	1	0.072 (0.012)	0.128 (0.013)	0.238 (0.020)	0.301 (0.023)
Gumbel (0.5, 2)	0.25	0.061 (0.009)	0.034 (0.008)	0.138 (0.016)	0.054 (0.010)
	0.5	0.074 (0.012)	0.039 (0.009)	0.206 (0.019)	0.087 (0.013)
	1	0.128 (0.015)	0.122 (0.014)	0.354 (0.022)	0.301 (0.022)
Normal distribution	0.25	0.058 (0.009)	0.025 (0.007)	0.103 (0.013)	0.032 (0.009)
	0.5	0.060 (0.010)	0.026 (0.007)	0.128 (0.015)	0.045 (0.010)
	1	0.049 (0.009)	0.114 (0.014)	0.160 (0.017)	0.288 (0.022)

Table 6 Empirical percentage of rejection of the symmetry test for weight parameter a and sample size n (with median δ estimated)

n	a	Normal ECF	Normal IECF	$N+\chi^2$ ECF	$N+\chi^2$ IECF
10	0.25	0.097 (0.013)	0.079 (0.013)	0.104 (0.014)	0.099 (0.013)
	0.5	0.098 (0.010)	0.061 (0.010)	0.109 (0.010)	0.080 (0.010)
	1	0.093 (0.012)	0.090 (0.011)	0.110 (0.015)	0.081 (0.013)
	1.5	0.093 (0.012)	0.096 (0.012)	0.108 (0.014)	0.095 (0.012)
	2	0.095 (0.013)	0.096 (0.014)	0.105 (0.014)	0.102 (0.013)
	3	0.096 (0.014)	0.095 (0.013)	0.105 (0.013)	0.107 (0.014)
	5	0.096 (0.013)	0.094 (0.013)	0.106 (0.013)	0.107 (0.013)
20	0.25	0.096 (0.012)	0.081 (0.011)	0.121 (0.016)	0.121 (0.014)
	0.5	0.095 (0.012)	0.074 (0.010)	0.129 (0.016)	0.120 (0.015)
	1	0.093 (0.015)	0.096 (0.012)	0.141 (0.016)	0.100 (0.015)
	1.5	0.096 (0.013)	0.100 (0.013)	0.145 (0.016)	0.108 (0.014)
	2	0.098 (0.014)	0.100 (0.013)	0.146 (0.016)	0.123 (0.014)
	3	0.100 (0.013)	0.099 (0.013)	0.143 (0.017)	0.140 (0.015)
	5	0.100 (0.013)	0.098 (0.013)	0.144 (0.015)	0.149 (0.016)

Table 7 Empirical percentage of rejection of the symmetry test for weight parameter a and sample size n (with median δ fixed)

n	a	Normal ECF	Normal IECF	$N+\chi^2$ ECF	$N+\chi^2$ IECF
10	0.25	0.051 (0.009)	0.051 (0.010)	0.088 (0.013)	0.087 (0.014)
	0.5	0.052 (0.010)	0.051 (0.009)	0.096 (0.014)	0.092 (0.014)
	1	0.052 (0.010)	0.053 (0.009)	0.098 (0.014)	0.084 (0.012)
	1.5	0.052 (0.010)	0.053 (0.009)	0.095 (0.013)	0.076 (0.012)
	2	0.052 (0.010)	0.053 (0.009)	0.090 (0.013)	0.071 (0.011)
	3	0.053 (0.009)	0.053 (0.009)	0.081 (0.012)	0.068 (0.011)
	5	0.052 (0.010)	0.053 (0.009)	0.072 (0.011)	0.068 (0.011)
20	0.25	0.051(0.009)	0.052(0.009)	0.137 (0.014)	0.128 (0.015)
	0.5	0.052 (0.010)	0.052 (0.010)	0.154 (0.014)	0.142 (0.016)
	1	0.052 (0.015)	0.052 (0.010)	0.161 (0.015)	0.127 (0.014)
	1.5	0.052 (0.010)	0.053 (0.010)	0.154 (0.014)	0.105 (0.013)
	2	0.052 (0.010)	0.052 (0.010)	0.142 (0.014)	0.090 (0.012)
	3	0.053 (0.009)	0.053 (0.010)	0.119 (0.013)	0.077 (0.011)
	5	0.052 (0.010)	0.053 (0.010)	0.090 (0.012)	0.068 (0.010)

empirical sizes and powers. For both tests the nominal size is fairly accurately recovered. Under alternatives, however, the power of the IECF test is overall higher than that of the ECF test in the case of the two-sample problem under normality, while being mostly less powerful (or with comparable performance) in other cases of the two-sample problem.

5.4 Tests for independence

For this problem we perform a permutation test as follows. Conditionally on the observed values $X_1, X_2, \ldots, X_n$, and $Y_1, Y_2, \ldots, Y_n$, of the two samples, we generate a number of permutations of the sample $X_1, X_2, \ldots, X_n$ and $Y_1, Y_2, \ldots, Y_n$, chosen at random from all possible permutations of each group, and proceed analogously as in the case of the two-sample problem. More detailed description of the procedure can be found in Székely et al. (2007). We compare our test $V_{n,a}$ in (3.17) for the null hypothesis η_0 with the test statistic

$$\begin{aligned}\widetilde{V}_{n,a} &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |\varphi_n(t_1,t_2) - \varphi_{X,n}(t_1)\varphi_{Y,n}(t_2)|^2 \mathrm{e}^{-a(|t_1|+|t_2|)}\mathrm{d}t_2\mathrm{d}t_1 \\ &= \frac{a^2}{n^2}\sum_{j,k=1}^{n} \frac{1}{(a^2+(X_j-X_k)^2)(a^2+(Y_j-Y_k)^2)} \\ &\quad + \frac{a^2}{n^4}\sum_{j,k,l,m=1}^{n} \frac{1}{(a^2+(X_j-X_l)^2)(a^2+(Y_k-Y_m)^2)} \\ &\quad - \frac{2a^2}{n^3}\sum_{j,k,l=1}^{n} \frac{1}{(a^2+(X_j-X_l)^2)(a^2+(Y_k-Y_l)^2)}.\end{aligned}$$

Table 12 presents the empirical sizes for sample size $n = 10$ under the null hypothesis of independence with both tests slightly overshooting the nominal size. Here (X, Y) are independent with marginal distributions, respectively, standard normal and Cauchy. For the ECF test, the results are obtained from 500 repetitions using the warp-speed bootstrap method [cf. Giacomini et al. (2013)]. Table 12 also presents the power of the two tests when (X, Y) follows a bivariate normal distribution with means zero, unit variances, and correlation 0.5. The figures show that the IECF test performs somewhat better than the ECF test and that for both tests the power increases as the weight parameter a gets larger.

6 Conclusion and outlook

We introduce the integrated empirical characteristic function (IECF) and suggest testing procedures for classical problems such as goodness of fit for parametric distributions, testing for symmetry and independence, as well as testing for the two-sample problem. The asymptotic distribution of the test statistic is obtained in each case, while

Table 8 Empirical percentage of rejection of the two-sample test under a null normal distribution for weight parameter a and sample size $n_1 = n_2 = 22$

a	Sample 2	ECF	IECF	Sample 2	ECF	IECF
0.25	Normal	0.053 (0.011)	0.051 (0.011)	Lognormal	0.820 (0.017)	0.997 (0.003)
0.50		0.053 (0.010)	0.050 (0.011)		0.913 (0.013)	0.997(0.002)
1.00		0.053 (0.010)	0.050 (0.011)		0.961 (0.009)	0.997(0.002)
1.50		0.052 (0.010)	0.050 (0.011)		0.975 (0.007)	0.997(0.002)
2.00		0.052 (0.011)	0.050 (0.011)		0.981 (0.007)	0.998(0.002)
0.25	Exponential	0.744 (0.019)	0.966 (0.008)	$U(0, 1)$	0.983 (0.006)	0.428 (0.024)
0.50		0.831 (0.016)	0.954 (0.009)		0.993 (0.004)	0.326 (0.022)
1.00		0.886 (0.014)	0.941 (0.010)		0.996 (0.003)	0.244 (0.020)
1.50		0.905 (0.012)	0.943 (0.010)		0.994 (0.003)	0.243 (0.020)
2.00		0.913 (0.012)	0.948 (0.010)		0.986 (0.006)	0.272 (0.021)
0.25	Gamma (2)	0.963 (0.008)	0.995 (0.003)			
0.50		0.992 (0.004)	1.000 (0.001)			
1.00		0.998 (0.002)	1.000 (0.000)			
1.50		1.000 (0.001)	1.000 (0.000)			
2.00		1.000 (0.001)	0.999 (0.001)			

Table 9 Empirical percentage of rejection of the two-sample test under a null exponential distribution for weight parameter a and sample size $n_1 = n_2 = 22$

a	Sample 2	ECF	IECF	Sample 2	ECF	IECF
0.25	Exponential	0.053 (0.010)	0.051 (0.010)	$L(0, 1)$	0.823 (0.019)	0.667 (0.021)
0.50		0.053 (0.010)	0.052 (0.010)		0.881 (0.015)	0.611 (0.021)
1.00		0.052 (0.010)	0.052 (0.010)		0.909 (0.013)	0.520 (0.022)
1.50		0.051 (0.010)	0.053 (0.010)		0.912 (0.014)	0.477 (0.023)
2.00		0.051 (0.010)	0.053 (0.010)		0.905 (0.014)	0.462 (0.022)
0.25	Gamma (2)	0.652 (0.022)	0.010 (0.004)	Lognormal	0.146 (0.016)	0.103 (0.014)
0.50		0.714 (0.021)	0.150 (0.016)		0.149 (0.017)	0.217 (0.017)
1.00		0.758 (0.020)	0.535 (0.024)		0.161 (0.017)	0.280 (0.018)
1.50		0.775 (0.018)	0.775 (0.018)		0.175 (0.018)	0.279 (0.019)
2.00		0.785 (0.019)	0.307 (0.022)		0.187 (0.017)	0.263 (0.020)

Table 10 Empirical percentage of rejection of the two-sample test under a null Gamma distribution for weight parameter a and sample size $n_1 = n_2 = 22$

a	Sample 2	ECF	IECF	Sample 2	ECF	IECF
0.25	Gamma (2)	0.051 (0.010)	0.052 (0.008)	Lognormal	0.322 (0.022)	0.220 (0.017)
0.50		0.050 (0.009)	0.052 (0.009)		0.368 (0.020)	0.040 (0.009)
1.00		0.049 (0.009)	0.051 (0.010)		0.393 (0.021)	0.094 (0.012)
1.50		0.050 (0.010)	0.051 (0.010)		0.393 (0.021)	0.101 (0.012)
2.00		0.050 (0.011)	0.050 (0.010)		0.384 (0.021)	0.084 (0.012)
0.25	Beta (2, 3)	1.000(0.000)	0.000(0.000)	$L(0, 1)$	0.819(0.015)	0.942(0.011)
0.50		1.000 (0.000)	0.062 (0.011)		0.916 (0.013)	0.946 (0.010)
1.00		1.000 (0.000)	0.780 (0.019)		0.966 (0.008)	0.942 (0.011)
1.50		1.000 (0.000)	0.553 (0.023)		0.979 (0.007)	0.939 (0.012)
2.00		1.000 (0.000)	0.343 (0.020)		0.985 (0.006)	0.939 (0.011)

Table 11 Empirical percentage of rejection of the two-sample test under a null logistic or lognormal distribution for weight parameter a and sample size $n_1 = n_2 = 22$

Sample 1	a	Sample 2	ECF	IECF	Sample 2	ECF	IECF
Logistic null	0.25	$L(0, 1)$	0.051 (0.009)	0.050 (0.010)	Lognormal	0.776 (0.019)	0.916 (0.013)
	0.50		0.050 (0.010)	0.050 (0.010)		0.855 (0.017)	0.908 (0.013)
	1.00		0.051 (0.010)	0.051 (0.010)		0.899 (0.014)	0.883 (0.014)
	1.50		0.051 (0.011)	0.051 (0.010)		0.911 (0.014)	0.866 (0.013)
	2.00		0.051 (0.011)	0.051 (0.009)		0.914 (0.013)	0.860 (0.014)
Lognormal null	0.25	Lognormal	0.053 (0.010)	0.051(0.012)	$U(0, 1)$	0.815 (0.018)	0.002 (0.002)
	0.50		0.052 (0.011)	0.050 (0.011)		0.920 (0.012)	0.100 (0.013)
	1.00		0.052 (0.011)	0.050(0.010)		0.972 (0.008)	0.613 (0.024)
	1.50		0.052 (0.010)	0.051 (0.011)		0.983 (0.006)	0.615 (0.025)
	2.00		0.051 (0.010)	0.051 (0.010)		0.986 (0.006)	0.538 (0.026)

Table 12 Empirical percentage of rejection of the independence test under a null (normal, Cauchy) independent components distribution and against a bivariate normal distribution with correlation 0.50 for weight parameter a and sample size $n = 10$

a	Size		Power	
	ECF	IECF	ECF	IECF
0.5	0.055 (0.018)	0.077 (0.020)	0.151 (0.033)	0.150 (0.028)
1.0	0.057 (0.018)	0.069 (0.018)	0.213 (0.029)	0.220 (0.038)
1.5	0.058 (0.018)	0.066 (0.018)	0.246 (0.033)	0.271 (0.039)
2.0	0.057 (0.016)	0.065 (0.017)	0.272 (0.038)	0.300 (0.041)
3.0	0.062 (0.017)	0.067 (0.017)	0.298 (0.038)	0.312 (0.040)
5.0	0.066 (0.018)	0.069 (0.017)	0.313 (0.040)	0.316 (0.038)

in practice the tests are implemented by using well known Monte Carlo and other resampling techniques. The tests so implemented compare well with more established procedures, such as those based on the ordinary empirical characteristic function. In particular, the IECF test appears to outperform the ECF test in certain cases of testing homogeneity, symmetry and independence, with the power margin being slight for some alternatives, while being more significant for other alternatives. Hence, in retrospect it appears that although integration, much like averaging, entails some loss of information, at the same time it provides cumulative information about the underlying law, so that in some cases the net result for the IECF seems to be positive in the sense that IECF procedures are more informative than their ECF-based counterparts. Clearly though this is an overall assessment, while a more educated inquiry into the specifics of this goes beyond the scope of the current work.

In closing and as the results obtained show promise, we note that the notion of a multivariate IECF and corresponding multivariate extension of the inferential procedures studied herein should be interesting research projects to be considered in the future.

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Appendix

In this "Appendix," we prove Propositions 2.1, 2.2 and 2.3. (We actually prove the 'only if' part of the propositions as the proof of the 'if' part is straightforward.) We also prove Theorem 4.1. Theorems 4.2, 4.3 and 4.4 can be proven following essentially the same lines as those presented for Theorem 4.1.

In what follows we often use the notation $\Phi_{\varphi_X,w}(t)$ for the ICF, and in this notation we write φ instead of φ_X when there is no confusion. In this connection we will also make use of the conjugation property of the ICF that $\overline{\Phi}_{\varphi,w}(t) = \Phi_{\overline{\varphi},w}(t)$.

Proof of Proposition 2.1. Assume $\Phi_{X,w}(t) = \Phi_{Y,w}(t), \forall t > 0$, and recall that the weight function $w(\cdot)$ is assumed to be continuous and that (any) CF is uniformly continuous for each $t \in \mathbb{R}$. By the standing assumptions on the weight function, we also have

$$\left|\int_0^\infty \varphi_X(u)w(u)\mathrm{d}u\right| \leq \int_0^\infty |\varphi_X(u)|\, w(u)\mathrm{d}u \leq \int_0^\infty w(u)\mathrm{d}u < \infty,$$

and likewise for the integral involving $\varphi_Y(\cdot)$. Hence, the identity

$$\int_0^t \varphi_X(u)w(u)\mathrm{d}u = \int_0^t \varphi_Y(u)w(u)\mathrm{d}u,$$

may be differentiated at each $t > 0$ to yield

$$\varphi_X(t)w(t) = \varphi_Y(t)w(t),$$

and since $w > 0$ we have

$$\varphi_X(t) = \varphi_Y(t), \tag{A.1}$$

$\forall t > 0$. Now for $t > 0$ consider

$$\begin{aligned}\Phi_{\varphi,w}(-t) &:= \int_0^{-t} \varphi(u)w(u)\mathrm{d}u = -\int_{-t}^0 \varphi(u)w(u)\mathrm{d}u = -\int_0^t \overline{\varphi(u)}w(u)\mathrm{d}u \\ &= -\Phi_{\overline{\varphi},w}(t) = -\overline{\Phi}_{\varphi,w}(t),\end{aligned}$$

where the third equality in the equation above follows by a simple change of variables in the integral and by the fact that $\varphi(-t) = \overline{\varphi(t)}$. Hence, if $\Phi_{X,w}(t) = \Phi_{Y,w}(t)$ holds for each $t > 0$, then the preceding argument implies that the same equality holds for each $t < 0$, which by repeating the steps of the proof entails that (A.1) is also true $\forall t < 0$, and the uniqueness of the CF completes the proof. □

Proof of Proposition 2.2. The assumption that the ICF is real entails that the imaginary part of the ICF satisfies $S_{X,w}(t) = 0,\ \forall t > 0$. As in the proof of Proposition 2.1, this implies that the imaginary part of the CF satisfies

$$S_X(t) = 0, \tag{A.2}$$

$\forall t > 0$, by simple differentiation.

Now for $t > 0$ consider $S_{X,w}(-t) := \int_0^{-t} S_X(u)w(u)\mathrm{d}u$, recall that $S_X(-t) = -S_X(t)$, and argue analogously as in the proof of Proposition 2.1 to conclude that $S_{X,w}(-t) = S_{X,w}(t)$. Hence, if $S_{X,w}(t) = 0$, holds $\forall t > 0$ then the same is true $\forall t < 0$, which implies that (A.2) holds for all $t \in \mathbb{R}$. However, this is true if and only if X is symmetrically distributed around zero (see Ushakov 1999, Theorem 1.1.8), and the proof is complete. □

Proof of Proposition 2.3. Consider first $t_1, t_2 > 0$, and recall again that the weight function $w(\cdot)$ is assumed to be continuous and that (any) CF is uniformly continuous. By the standing assumptions on the weight function, we also have

$$\left|\int_0^\infty \int_0^\infty \varphi_{X,Y}(u,v)w(u)w(v)\mathrm{d}v\mathrm{d}u\right| \leq \int_0^\infty \int_0^\infty \left|\varphi_{X,Y}(u,v)\right| w(v)w(u)\mathrm{d}v\mathrm{d}u$$
$$\leq \left(\int_0^\infty w(u)\mathrm{d}u\right)^2 < \infty.$$

Hence, the partial derivative of the ICF

$$\Phi_w(t_1,t_2) = \int_0^{t_1}\int_0^{t_2} \varphi(u_1,u_2)w(u_2)w(u_1)\mathrm{d}u_2\mathrm{d}u_1, \tag{A.3}$$

of second order may be computed from (A.3) as

$$\frac{\partial^2\Phi_w(t_1,t_2)}{\partial t_1\partial t_2} = \varphi(t_1,t_2)w(t_1)w(t_2), \tag{A.4}$$

irrespective of the order of differentiation. Assuming that (2.8) holds and arguing as in Proposition 2.1, the partial derivative figuring in the left-hand side of (A.4) may equivalently be computed by differentiating the right-hand side of (2.8) as

$$\frac{\partial^2\Phi_w(t_1,t_2)}{\partial t_1\partial t_2} = \varphi_X(t_1)\varphi_Y(t_2)w(t_1)w(t_2). \tag{A.5}$$

Equating now the right-hand sides of (A.4) and (A.5), we get

$$\varphi(t_1,t_2) = \varphi_X(t_1)\varphi_Y(t_2), \tag{A.6}$$

$\forall t_1, t_2 > 0$. Now for $t_1, t_2 > 0$ consider

$$\begin{aligned}\Phi_{\varphi,w}(-t_1,-t_2) &:= \int_0^{-t_1}\int_0^{-t_2} \varphi(u_1,u_2)w(u_2)w(u_1)\mathrm{d}u_2\mathrm{d}u_1\\ &= \int_{-t_1}^{0}\int_{-t_2}^{0} \varphi(u_1,u_2)w(u_2)w(u_1)\mathrm{d}u_2\mathrm{d}u_1\\ &= \int_0^{t_1}\int_0^{t_2} \overline{\varphi(u_1,u_2)}w(u_2)w(u_1)\mathrm{d}u_2\mathrm{d}u_1 = \Phi_{\overline{\varphi},w}(t_1,t_2)\\ &= \overline{\Phi}_{\varphi,w}(t_1,t_2),\end{aligned}$$

where the third equality in the equation above follows by a simple change of variables in the integral and the fact that $\varphi(-t_1,-t_2) = \overline{\varphi(t_1,t_2)}$. Using the last equation and recalling that in the univariate case $\Phi_{\varphi,w}(-t) = -\overline{\Phi}_{\varphi,w}(t)$, we conclude that if (2.8) is true for each pair $t_1, t_2 > 0$, the same holds for each pair $t_1, t_2 < 0$, which by repeating the steps of the proof entails that (A.6) is also true $\forall t_1, t_2 < 0$.

As shown above the validity of (2.8) $\forall t_1, t_2 > 0$ was sufficient to conclude that (A.6) holds not only $\forall t_1, t_2 > 0$, but also $\forall t_1, t_2 < 0$. The same argument, however, cannot be extended to cover the case $t_1 < 0, t_2 > 0$, and therefore, one needs to assume that (2.8) also holds for values of (t_1, t_2) in the second quadrant and repeat the steps leading to (A.6). The case of the forth quadrant $(t_1 > 0, t_2 < 0)$ follows by the conjugation property of the CF, thereby completing the proof. □

Proof of Theorem 4.1. Let $Z_n(t) = \sqrt{n}(\hat{\varphi}_n(t) - \varphi_0(t))$ and $\mathcal{Z}_1$ is a zero-mean Gaussian process with covariance being the same as $w(s, t)$ in Meintanis and Swanepoel (2007). It can be seen that for all $m \geq 1$,

$$Z_n \xrightarrow{w} \mathcal{Z}_1 \text{ on } C[0, m],$$

where $C[0, m]$ denotes the set of all continuous complex-valued functions defined on $[0, m]$ [cf. Theorem 12.3 of Billingsley (1968)]. Put

$$Y_n = \int_0^\infty \left| \int_0^t Z_n(u) w(u) \mathrm{d}u \right|^2 W(t) \mathrm{d}t \quad \text{and} \quad Y_{n,m} = \int_0^m \left| \int_0^t Z_n(u) w(u) \mathrm{d}u \right|^2 W(t) \mathrm{d}t.$$

By the continuous mapping theorem, we have that as $n \to \infty$,

$$Y_{n,m} \xrightarrow{d} Y_m := \int_0^m \left| \int_0^t \mathcal{Z}_1(u) w(u) \mathrm{d}u \right|^2 W(t) \mathrm{d}t,$$

and further, by the monotone convergence theorem, as $m \to \infty$,

$$Y_m \xrightarrow{d} Y := \int_0^\infty \left| \int_0^t \mathcal{Z}_1(u) w(u) \mathrm{d}u \right|^2 W(t) \mathrm{d}t.$$

To assert the theorem, it remains to show that for all $\epsilon > 0$,

$$\lim_{m \to \infty} \limsup_{n \to \infty} P(|Y_n - Y_{n,m}| \geq \epsilon) = 0 \tag{A.7}$$

[cf. Proposition 6.3.9 of Brockwell and Davis (1987)]. It is easy to show that

$$\sup_{n,u} E|Z_n(u)|^2 < \infty. \tag{A.8}$$

Then, using the Schwarz inequality and (A.8), we get

$$\begin{aligned} E|Y_n - Y_{n,m}| &\leq \int_m^\infty E \left\{ \int_0^t |Z_n(u)|^2 w(u) \mathrm{d}u \int_0^t w(u) \mathrm{d}u \right\} W(t) \mathrm{d}t \\ &\leq C \int_m^\infty W(u) \mathrm{d}u \quad \text{for some } C > 0, \end{aligned}$$

which indicates (A.7). This establishes the theorem. □

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